

APPROXIMATE DISTRIBUTIONS FOR THE BOX CLASS OF LIKELIHOOD CRITERIA

N.A.S. Crowther

Department of Math. Statistics, University of Port Elizabeth, P.O. Box 1600, Port Elizabeth, Rep. of South Africa, 6000

Key words: Beta distribution; Box series; percentage points; sphericity test; Wilks criterion

SUMMARY

Alternative Box-type expansions for the distribution functions of a class of likelihood criteria are presented. These converge in general more rapidly than the standard Box procedure and simplify applications.

1. Introduction

The large sample distribution theory for a class of likelihood criteria developed by Box (1949) is well known and extensively used in many applications. Exact distributions for some of these criteria have been provided by various authors and corresponding tables of percentage points were presented. Using these exact distributions involve tedious computations and in general it is not feasible to provide a complete set of tables.

We consider three alternative Box-type expansions. The first of which is easier to apply than the standard Box procedure. The second is in general more accurate and the third has both of these properties. It is also shown that the distribution of a member of the class may in general, even for moderate sample sizes, be represented sufficiently accurate by a function of a single beta random variable with explicit parameters. Percentage points may then be computed directly.

We illustrate the results in the special cases of the Wilks criterion and the sphericity test criterion. Extensions to other criteria in the class are straightforward and will be omitted.

2. Notation

We shall make use of the following notation:

$B_{\alpha,\beta}$: a beta distribution with parameters α and β .

$B_{\alpha,\beta}(x)$: a beta distribution function with parameters α and β .

χ^2_ν : a chi-square distribution with ν degrees of freedom.

$C_\nu(x)$: a chi-square distribution function with ν degrees of freedom.

$c_\nu(x)$: a chi-square density function with ν degrees of freedom.

$O(\rho^{-k})$: a term of order ρ^{-k} . (If $h(\rho) = O(\rho^{-k})$, then $|\rho^k h(\rho)|$ is bounded as $|\rho| \rightarrow \infty$.)

3. The Box Class of Likelihood Criteria

Box (1949) considered a class of random variables w , where the characteristic function of $-2\rho \ln w$ is given by:

$$(3.1) \quad \Phi(t) = K \frac{\left[\prod_{j=1}^b y_j^{y_j} \right]^{-2\rho it}}{\left[\prod_{k=1}^a x_k^{x_k} \right]} \frac{\prod_{k=1}^a \Gamma(x_k(1-2\rho it) + \xi_k)}{\prod_{j=1}^b \Gamma(y_j(1-2\rho it) + \eta_j)}$$

for some constant K and $\sum x_k = \sum y_j$. In applications the constants may be chosen in such a way that ρ grows with the sample size.

Suppose that $\Phi(t)$ can be represented in the series form:

$$(3.2) \quad \Phi(t) = \Phi_v(t) + \phi_1(t) \sum_{r=0}^{\infty} \mu_r \phi^r(t)$$

where the $\{\mu_k\}$ are constants, $\Phi_v(t)$ the characteristic function of some random variable v and $\phi_1(t)$ and $\phi(t)$ functions for which the corresponding inverse

$$(3.3) \quad f_r(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi_1(t) \phi^r(t) dt,$$

exists and is known. Assuming convergence, the density corresponding to $\Phi(t)$ may then be represented as

$$(3.4) \quad f(x) = f_v(x) + \sum_{r=0}^{\infty} \mu_r f_r(x)$$

where $f_v(x)$ is the density of v .

For the corresponding distribution functions it holds that:

$$(3.5) \quad F(x) = F_v(x) + \sum_{r=0}^{\infty} \mu_r F_r(x).$$

It follows from (3.2) that:

$$(3.6) \quad \{\phi_1(t)\}^{-1} \{\Phi(t) - \Phi_v(t)\} = \sum_{r=0}^{\infty} \mu_r \phi^r(t) = g(\phi(t)), \text{ say.}$$

The LHS of (3.6) should therefore be only a function of $\phi(t)$. Let $\theta = \phi(t)$. Then:

$$(3.7) \quad g(\theta) = \{ \phi_1(\phi^{-1}(\theta)) \}^{-1} \{ \Phi(\phi^{-1}(\theta)) - \Phi_v(\phi^{-1}(\theta)) \} = \sum_{r=0}^{\infty} \mu_r \theta^r.$$

Hence the coefficients $\{\mu_r\}$ are the coefficients of θ^k in the Maclaurin series expansion of $g(\theta)$.

The procedure we follow is choosing a convenient approximation $F_v(x)$, such that the contribution of the other terms in the RHS of (3.5) is relatively small. We consider four types of representations of $F(x)$, first of which is the well-known Box expansion.

3.1 The Box expansion.

In this case v is a χ^2 random variable and $F_r(x)$ χ^2 distribution functions, i.e.,

$$\Phi_v(t) = (1-2it)^{-\gamma/2}$$

$$\theta = \phi(t) = (1-2it)^{-1} \text{ or } 2it = 1 - \theta^{-1}$$

$$\theta_1(t) = (1-2it)^{-\gamma/2}$$

and γ is determined by (3.8). From Box (1949) (see Anderson (1958), Section 8.6) it follows that

$$(3.8) \quad \Phi\left(\frac{1}{2i}(1-\theta^{-1})\right) = K \left[\frac{\prod_{j=1}^b y_j}{\prod_{k=1}^a x_k} \right]^{\rho(\theta^{-1}-1)} \frac{\prod_k (x_k \rho \theta^{-1} + x_k(1-\rho) + \xi_k)}{\prod_j (y_j \rho \theta^{-1} + y_j(1-\rho) + \eta_j)}$$

$$= \theta^{1/2\gamma} \exp \left\{ \sum_{r=1}^{\infty} \omega_r (\theta^r - 1) \right\}$$

where $\gamma = -2 \{ \sum \xi_k - \sum \eta_j - \frac{1}{2}(a-b) \}$

$$\omega_r = \frac{(-1)^{r+1}}{\rho^r r(r+1)} \left\{ \sum_k B_{r+1}(\beta_k + \xi_k) / x_k^r - \sum_j B_{r+1}(\epsilon_j + \eta_j) / y_j^r \right\}$$

$\beta_k = (1-\rho)x_k$, $\epsilon_j = (1-\rho)y_j$ and $B_{r+1}(\cdot)$ is a Bernoulli polynomial. In general ρ is chosen to make at least $\omega_1 = 0$ and $\omega_r = O(\rho^{-r})$. Including terms of order up to ρ^{-k} , it follows that:

$$(3.9) \quad g(\theta) = \exp \left\{ \sum_{r=2}^k \omega_r (\theta^r - 1) \right\} - 1 = \sum_{r=0}^k \mu_r \theta^r$$

where $\mu_1 = 0$, $\mu_0 = O(\rho^{-2})$ and $\mu_r = O(\rho^{-r})$ for $r \geq 2$. Hence up to $O(\rho^{-k})$:

$$(3.10) \quad F(x) = C_\gamma(x) + \sum_{r=0}^k \mu_r C_{\gamma+2r}(x)$$

By putting $\theta = 1$ in (3.9) it follows that

$$(3.11) \quad \sum_{r=0}^k \mu_r = 0.$$

3.2 *via χ^2 variable and $\{F_r(x)\}$ χ^2 densities*

Let $\Phi_v(t)$ and $\phi(t)$ be the same as in 3.1, and

$$\phi_1(t) = -it(1-2it)^{-\gamma/2-1}.$$

Then:

$$(3.12) \quad F_r(x) = -\frac{1}{2\pi} \int_{-\infty}^x \int_{-\infty}^{\infty} e^{-itz} it(1-2it)^{-\gamma/2-1} dt dz$$

$$= \frac{1}{\Gamma(\gamma/2+r)2^{\gamma/2+r}} e^{-\frac{1}{2}x} x^{\frac{1}{2}\gamma+r}$$

which can be seen by interchanging the order of integration. Including terms up to order ρ^{-k} :

$$(3.13) \quad g(\theta) = \frac{2}{1-\theta} \sum_{r=0}^k \mu_r \theta^r = \sum_{r=0}^k \mu'_r \theta^r$$

where μ_r is given by (3.9) and

$$(3.14) \quad \mu'_r = 2 \sum_{j=0}^r \mu_j.$$

From (3.11) and (3.14) we have that

$$(3.15) \quad \mu'_r = -2 \sum_{j=r+1}^k \mu_j$$

which proves that $\mu'_r = O(\rho^{-r-1})$ for $r \geq 1$, $\mu'_0 = O(\rho^{-2})$ while $\mu'_k = 0$. It follows that, including terms of order ρ^{-k} :

$$(3.16) \quad F(x) = C_\gamma(x) + \sum_{r=0}^{k-1} \mu'_r C_{\gamma+2r+2}(x).$$

The summation in (3.16) is in terms of χ^2 densities, which is a simpler form of

representation than (3.10). For the same k , (3.16) and (3.10) will give the same answer.

3.3 *v* a function of a beta variable and $\{F_r(x)\}$ χ^2 distribution functions

Let $v = -2\rho_0 \ln w_0$, where w_0 has a $B_{\alpha,\beta}$ distribution. Hence:

$$\phi_v(t) = K_0 \frac{\Gamma(\alpha - 2\rho_0 it)}{\Gamma(\alpha + \beta - 2\rho_0 it)}$$

for some constant K_0 . The choices for $\phi(t)$ and $\phi_1(t)$ are the same as in Section 3.1, i.e.:

$$\theta = \phi(t) = (1 - 2it)^{-1} \quad \text{and} \quad \phi_1(t) = (1 - 2it)^{-\gamma/2}.$$

Similarly as in (3.8) we obtain that:

$$(3.17) \quad \phi_v\left(\frac{1}{2i}(1 - \theta^{-1})\right) = \theta^\beta \exp \left\{ \sum_{r=1}^{\infty} \psi_r(\theta^r - 1) \right\}$$

where

$$\psi_r = \frac{(-1)^r}{\rho_0^r r(r+1)} \{B_{r+1}(\alpha + \beta - \rho_0) - B_{r+1}(\alpha - \rho_0)\}.$$

Hence:

$$(3.18) \quad \beta = \frac{1}{2}\gamma$$

and if

$$(3.19) \quad \rho_0 = \alpha + \frac{1}{2}(\beta - 1)$$

then

$$(3.20) \quad \psi_r = \begin{cases} \frac{2}{\rho_0^r r(r+1)} B_{r+1}(\frac{1}{2}(\beta + 1)), & r \text{ even} \\ 0, & r \text{ odd.} \end{cases}$$

It then follows that up to $O(\rho^{-k}) + O(\rho_0^{-k})$:

$$(3.21) \quad g(\theta) = \exp \left\{ \sum_{r=2}^k \omega_r(\theta^r - 1) \right\} - \exp \left\{ \sum_{r=2}^k \psi_r(\theta^r - 1) \right\} \\ = \sum_{r=0}^k \mu_r \theta^r.$$

By putting $\theta = 1$ in (3.20), it again follows that:

$$(3.22) \quad \sum_{r=0}^k \mu_r = 0$$

where $\mu_r = O(\rho^{-r}) + O(\rho_0^{-r})$, $\mu_1 = 0$ and it follows that $\mu_0 = -\sum_{r=2}^k \mu_r$. We may now choose α in such a way that $\omega_2 = \psi_2$, i.e.

$$(3.23) \quad \rho_0 = \delta \rho,$$

where

$$(3.24) \quad \delta = [2B_3(\frac{1}{2}(\beta+1)) / \{ \sum_j B_3(\epsilon_j + \eta_j) / y_j^2 - \sum_k B_3(\beta_k + \xi_k) / x_k^2 \}]^{1/2}$$

In applications δ will be independent of the sample size while ρ and ρ_0 will be of the same order. This choice of α implies that $\mu_2 = 0$ up to order ρ^{-2} . We now have the following representation of $F(x)$ up to terms of order ρ^{-k} :

$$(3.25) \quad F(x) = F_v(x) + \sum_{r=0}^k \mu_r C_{\gamma+2r}(x)$$

In (3.25) $v = -2\rho_0 \ln w_0$ where w_0 has a $B_{\alpha,\beta}$ distribution, α and β being given by (3.19) and (3.18) respectively. Furthermore $\mu_1 = 0$, μ_0 and μ_2 are $O(\rho^{-3})$ and $\mu_r = O(\rho^{-r})$, $r \geq 3$.

Numerical experience has shown that the representation (3.25) is for the same k much more accurate than the Box expansion (3.10) (and (3.16)). Even for moderate sample sizes, neglecting $O(\rho^{-3})$, we may in general approximate $F(x)$ by $F_v(x)$ quite accurately. Hence $-2\rho \ln w$ is approximately distributed as $-2\rho_0 \ln w_0$ and we have the following result:

$$(3.26) \quad \text{The random variable } w \text{ is approximately distributed as } w_0^\delta, \text{ where } w_0 \text{ has a } B_{\alpha,\beta} \text{ distribution, the parameters } \alpha \text{ and } \beta \text{ being given by (3.19) and (3.18).}$$

Tukey and Wilks (1946) also approximated the distribution of w by the distribution of a power of a single beta random variable, using a method of equating moments.

3.4 v a function of a beta variable and $\{F_r(x)\}$ χ^2 densities

We finally choose the random variable v as in Section 3.3, but $\phi(t)$ and $\phi_1(t)$ as in Section 3.2. We then have that up to order ρ^{-k} :

$$(3.27) \quad F(x) = F_v(x) + \sum_{r=0}^{k-1} \mu'_r c_{\gamma+2r+2}(x)$$

where

$$(3.28) \quad \mu'_r = 2 \sum_{j=0}^r \mu_j = -2 \sum_{j=r+1}^k \mu_j,$$

the μ_j being given by (3.21). Here μ'_0 and μ'_1 are $O(\rho^{-3})$, while $\mu'_r = O(\rho^{-r-1})$.

The representation (3.27) is of the same accuracy as (3.25), but simpler to use since the series involves only χ^2 densities. This representation of $F(x)$ is therefore preferable, particularly for calculating percentage points.

4. Special Cases

4.1 The Wilks Likelihood Ratio Statistic $u_{p,m,n}$

Consider the statistic

$$(4.1) \quad u_{p,m,n} = \prod_{i=1}^p x_i$$

where the x_i are independently beta distributed with corresponding parameters $(n+1-i)/2$ and $m/2$

The exact distribution of $u_{p,m,n}$ had been studied by various authors and Schatzoff (1966) developed a representation of this distribution in closed form when p or m is even, using a method of successive convolutions. He also presented some tables of correction factors for converting χ^2 percentiles to percentiles of $-2\rho \ln u_{p,m,n}$. Lee (1972) extended this to the case where p and m are both odd. A problem with the Schatzoff procedure is that the number of significant digits which must be retained in intermediate calculations increases as p , m and n increase. Tretter and Walster (1975) obtained a representation of the distribution as a mixture of beta distributions. Lee, Krishnaiah and Chang (1977) fitted a Pearson Type 1 distribution, using the first four moments and presented some tables of percentage points.

The characteristic function of $-2\rho \ln u_{p,m,n}$ is given by (see Anderson (1958)):

$$(4.2) \quad \phi(t) = K \prod_{j=1}^p \frac{\Gamma(\frac{n+1-j}{2} - 2\rho it)}{\Gamma(\frac{n+m+1-j}{2} - 2\rho it)}.$$

In this case:

$$(4.3) \quad a = b = p, x_k = y_k = 1, \beta_k = 1 - \rho, \epsilon_j = 1 - \rho,$$

$$\xi_k = \frac{1}{2}(n-1-k), \eta_j = \frac{1}{2}(n+m-1-j), f = mp..$$

$$(4.4) \quad \omega_r = \frac{(-1)^r}{\rho^r r(r+1)} \sum_{j=1}^p \{B_{r+1} (\frac{1}{2}(n+m+1-j) - \rho) - B_{r+1} (\frac{1}{2}(n-p+j) - \rho)\}$$

If

$$(4.5) \quad \rho = \frac{1}{2}(2n+m-p-1)$$

then

$$(4.6) \quad \omega_r = \begin{cases} \frac{2}{\rho^r r(r+1)} \sum_{j=1}^p B_{r+1} (\frac{1}{2}(m+p-2j+3)), & r \text{ even} \\ 0 & , r \text{ odd} \end{cases}$$

Since

$$(4.7) \quad \beta = \frac{1}{2}mp$$

it follows from (3.24) and (4.3) that

$$(4.8) \quad \delta = \sqrt{(m^2 p^2 - 4)/(m^2 + p^2 - 5)}.$$

We have as a special case of (3.26) the result of Rao (1952):

(4.9) *The Wilks likelihood ratio statistic $u_{p,m,n}$ is approximately distributed as w_0^δ where w_0 has a beta distribution with parameters $\alpha = \rho_0 - \frac{1}{2}(\beta-1)$ and $\beta = \frac{1}{2}mp$, $\rho_0 = \delta\rho$, $\rho = \frac{1}{2}(2n+m-p-1)$ and $\delta = \sqrt{(m^2 p^2 - 4)/(m^2 + p^2 - 5)}$.*

Note that if $m = 1$ or $p = 1$, then $\delta = 1$. If $m = 2$ or $p = 2$, then $\delta = 2$ which agrees with the well-known result that in this case, $u_{p,m,n}$ is exactly distributed as the square of a beta random variable. The result (4.9) can be regarded as a generalization of this.

In general α is not an integer or half-integer and one has to interpolate in the standard beta tables. If α and β are both moderately large (≥ 4), Carter's formula is sufficiently accurate (see Pearson and Hartley (1966)).

Example 4.1

Consider $u_{3,4,7}$. In this case:

$$\rho = 3.5; \delta = \sqrt{7}; \rho_0 = 3.5\sqrt{7} = 9.26; \beta = 6; \alpha = 6.76.$$

The lower 5% point for $u_{3,4,7}$ is:

$$u_{0,05} = (0,305)\sqrt{7} = 0,0432$$

The corresponding χ^2 correction factor is $-2\rho \ln u_{0,05}/\chi_{0,05;12}^2 = 1,046$ which is the same as that given by Schatzoff (1966). Similarly $u_{0,01} = (0,225)\sqrt{7} = 0,01932$ and the χ^2 correction factor is 1,054.

A rough rule is that if $n \geq p+m$, percentage points obtained by using (4.9) are sufficiently accurate and agree with tables presented by Schatzoff (1966), Lee, Krishnaiah and Chang (1977) and others.

For using (3.27) we include the μ'_k up to $O(n^{-6})$, i.e. for $k = 6$:

$$\begin{aligned} \mu'_0 &= -2\omega_4 + 2\psi_4 - 2\omega_6 + 2\psi_6 + 2\omega_2\omega_4 - 2\psi_2\psi_4 \\ \mu'_1 &= \mu'_0 \\ \mu'_2 &= -2\omega_4 + 2\psi_4 - 2\omega_6 + 2\psi_6 \\ \mu'_3 &= \mu'_2 \\ \mu'_4 &= -2\omega_2\omega_4 + 2\psi_2\psi_4 - 2\omega_6 + 2\psi_6 \\ \mu'_5 &= \mu'_4 \end{aligned} \quad (4.10)$$

4.2 The Sphericity Test Criterion

The test statistic is given by

$$(4.11) \quad w = |S| / \{(\text{tr } S)/p\}^p$$

where S has a $W(1, n)$ distribution. (See Anderson (1958), Chapter 10.) The characteristic function of $-2\rho \ln w$ is given by:

$$(4.12) \quad \Phi(t) = K p^{-2\rho p i t} \left\{ \prod_{j=1}^p \Gamma(\frac{1}{2}(n-j+1) - 2\rho i t) \right\} / \Gamma(\frac{1}{2}np - 2\rho p i t).$$

(Note that ρ corresponds to ρx_k in Anderson (1958).)

In this case:

$$(4.13) \quad \rho = \frac{1}{2}n - (2p^2 + p + 2)/12p, \beta = \frac{1}{2}(p+2)(p-1)$$

$$\omega_2 = (p+2)(p-1)(p-2)(2p^3 + 6p^2 + 3p + 2)/1152p^2$$

$$\psi_2 = (p+2)(p-1)(p-2)(p+3)(p^2+p+2)/1536$$

$$\delta = \sqrt{\frac{1}{4}p^2(p+3)(p^2+p+2)/(2p^3+6p^2+3p+2)}$$

- (4.14) *The sphericity test criterion w is approximately distributed as w_0^δ , where w_0 has a beta distribution with parameters $\alpha = \rho_0 - \frac{1}{2}(\beta-1)$ and $\beta = \frac{1}{4}(p+2)(p-1)$, where $\rho_0 = \rho\delta$.*

If $p = 2$, then $\beta = 1$ and $\alpha = \frac{1}{2}(n-1)\sqrt{2/5}$, which agrees with the well-known fact that in this case, w is exactly beta distributed with parameters $\frac{1}{2}(n-1)$ and 1.

Nagarsenker and Pillai (1973) obtained representations for the exact distribution of w and provided tables for $p = 4(1)10$.

Example 4.2

If $p = 6$ and $n = 18$, then:

$$\alpha = 27.06; \beta = 10; \delta = 4.00$$

The lower 5% point for w is $(0.6049)^4 = 0.1339$ and the lower 1% point is $(0.5477)^4 = 0.0900$. The corresponding exact points given by Nagarsenker and Pillai are 0.1337 and 0.0898 respectively.

A rough rule is that if $n \geq 3p$, the percentage points obtained by using (4.14) are accurate to four decimal places. In order to illustrate that it is even fairly accurate in other cases, we compare in Table 1 some of the percentage points obtained with the exact percentage points given by Nagarsenker and Pillai (1973).

TABLE 1

	$p = 5, n = 6$		$p = 5, n = 13$	
	5%	1%	5%	1%
(4.14)	0,0 ² 152	0,0 ³ 298	0,11483	0,06938
Nagarsenker, Pillai	0,0 ² 12621	0,0 ³ 21839	0,11460	0,069151

ACKNOWLEDGEMENT

The author wishes to thank the referee for some useful comments.

REFERENCES

- ANDERSON, T.W. (1958). *An Introduction to Multivariate Statistical Analysis*. New York: Wiley.
- BOX, G.E.P. (1949). A general distribution theory for a class of likelihood criteria. *Biometrika* **36**, 317-346.
- DAVIS, A.W. (1971). Percentile approximations for a class of likelihood criteria. *Biometrika* **58**, 349-356.
- LEE, J.C., KRISHNAIAH, P.R. and CHANG, T.C. (1977). Approximations to the distributions of the determinants of real and complex multivariate beta matrices. *South African Statist. J.* **11**, 13-26.
- LEE, Y.S. (1972). Some results on the distribution of Wilks's likelihood-ratio criterion. *Biometrika* **59**, 649-664.
- NAGARSENKER, B.N. and PILLAI, K.C.S. (1973). The distribution of the sphericity test criterion. *J. Multivar. Anal.* **3**, 226-235.
- PEARSON, E.S. and HARTLEY, H.O. (1966). *Biometrika Tables for Statisticians*. London: Cambridge University Press.
- RAO, C.R. (1952). An asymptotic expansion of the distribution of Wilks' Λ criterion. *Bull. Int. Stat. Inst.* **33**(2), 177-180.
- SCHATZOFF, M. (1966). Exact distributions of Wilks's likelihood ratio criterion. *Biometrika* **53**, 347-358.
- TRETTTER, M.J. and WALSTER, G.W. (1975). Central and noncentral distribution of Wilks's statistic in MANOVA as mixtures of incomplete beta functions. *Ann. Statist.* **3**, 467-472.
- TUKEY, J.W. and WILKS, S.S. (1946). Approximation of the distribution of the product of beta variables by a single beta variable. *Ann. Math. Statist.* **17**, 318-324.